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Service is experiencing high load. You may experience slower response times.

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Introduction

SAP-RPT is a relational pretrained foundation model that delivers accurate predictive insights from structured business data. It uses in-context learning, allowing users to provide example records directly in API calls, to generate instant, reliable predictions without any model training.

Key Features

Pretrained Model
Eliminates costly and time-consuming model training

In-Context Learning
Instant results through example records at runtime

Data Resilient
Delivers dependable predictions even with changing data

Business-Grounded
Accurate, enterprise data-driven insights

Natural Language Interface
Ask questions in natural language via Chat-RPT

What-If Scenarios
Simulate changes and predict outcomes instantly

Uploading CSV Data

Supported File Format

SAP-RPT accepts CSV (Comma-Separated Values) files with the following specifications:

- File extension: .csv

- Maximum file size: 5MB
- First row should contain column headers
- Data should be properly formatted with consistent columns

Limitations

- **Maximum rows**: 2,073 rows per file
- **Maximum columns**: 50 columns per file
- **Maximum cell length**: 1,000 characters per cell
- **Maximum column name length**: 100 characters

Best Practices

- Ensure your data is clean and free of missing values where possible
- Use descriptive column names (under 100 characters)
- Keep consistent data types within each column
- Remove any unnecessary columns before upload

API Reference

Authentication

All API requests require authentication using your personal API token. See the API Token section for details on obtaining and using your token.

Endpoints

POST /api/predict

Submit data for prediction using in-context learning. The model learns from example rows (context rows) and predicts values for query rows that contain the `[PREDICT]` placeholder.

Request Body:

```
{
  "rows": [
```

```
    {
      "customer_id": "C001",
      "age": 35,
      "subscription_months": 12,
      "churn": "No"
    },
    {
      "customer_id": "C042",
      "age": 20,
      "subscription_months": 1,
      "churn": "Yes"
    },
    {
      "customer_id": "C002",
      "age": 28,
      "subscription_months": 3,
      "churn": "[PREDICT]"
    }
  ],
  "index_column": "customer_id"
}
```

Request Parameters:

- `rows` (required): Array of data rows. Each row is an object with column names as keys.
  - **Context rows**: Rows with complete data (no `[PREDICT]` values) used as examples
  - **Query rows**: Rows containing `[PREDICT]` placeholder values indicating what to predict
- `index_column` (optional): Column name to use as row identifier in the response

Constraints:

- Minimum 1 query row (rows with `[PREDICT]`)
- Maximum 50 query rows
- Minimum 2 context rows (rows without `[PREDICT]`)
- Maximum 2,048 context rows
- Maximum 4 target columns (columns containing `[PREDICT]`)
- Maximum 50 columns total per row

- Maximum 2,073 rows total
- Maximum 1,000 characters per cell
- Maximum 100 characters per column name

Response:


```json
{
  "prediction": {
    "id": "c5e39406-351a-484b-a668-18d8e14b5666",
    "metadata": {
      "num_columns": 3,
      "num_predict_rows": 1,
      "num_predict_tokens": 1,
      "num_rows": 2
    },
    "predictions": [
      {
        "churn": [
          {
            "confidence": null,
            "prediction": "Yes"
          }
        ],
        "customer_id": "C002"
      }
    ]
  },
  "delay": 1250.5
}
```


Response Fields:

- **`prediction`** : The prediction response object
  - **`id`** : Unique prediction identifier
  - **`predictions`** : Array of prediction results, one per query row
    - Each element of the array contains the predicted column(s) with a prediction object
    - When passing the optional **`index_column`** input parameter, each element of the array also contains the chosen column with its original input value (e.g. **`customer_id`** in the example above)

    - `metadata` : Summary information about the prediction
- `delay` : Processing time in milliseconds

Example: Predicting Multiple Columns

```
{
  "rows": [
    {
      "product": "Widget A",
      "price": 29.99,
      "quantity": "[PREDICT]",
      "revenue": "[PREDICT]"
    },
    {
      "product": "Widget B",
      "price": 25.95,
      "quantity": 2,
      "revenue": 10025.78
    },
    {
      "product": "Widget C",
      "price": 39.99,
      "quantity": 150,
      "revenue": 5998.50
    }
  ]
}
```

Error Codes

- **400**: Bad Request - Invalid data format or validation error
- **401**: Unauthorized - Invalid or missing API token
- **429**: Too Many Requests - Rate limit exceeded or service temporarily unavailable (includes `Retry-After` header)
- **503**: Service Unavailable - Server is under high load. Retry your request after the time indicated in the `Retry-After` header
- **500**: Internal Server Error - Contact support

Rate Limits

API requests are rate-limited to prevent abuse. If you exceed the limit, you'll receive a 429 status code with a `Retry-After` header indicating when you can retry.

Chat-RPT

Chat-RPT is a conversational natural language interface for SAP-RPT. Instead of manually constructing prediction requests, you can ask questions in any natural language and the system will automatically classify your intent, build the appropriate query, run predictions, and return results with explanations.

Capabilities

What-If Analysis

Simulate changes to your data and predict outcomes without modifying the original dataset. Supports both categorical changes (e.g., swap a supplier) and numeric adjustments (e.g., reduce cost by 20%).

Example queries:

What if I change the supplier to Acme Corp for all Tokyo orders, predict the delivery cost?

What if I reduce the quantity by 20%, how would CO2 emissions change?

Simulate switching the region to EMEA and predict revenue impact

Add Row & Predict

Add a new hypothetical record to your dataset and instantly get a prediction for the target column. The model uses your existing data as context to predict the new row's outcome.

Example queries:

I have a new order with destination Berlin, quantity 500 -- predict the CO2 emissions

Add a customer with age 25, subscription 2 months -- will they churn?

New equipment: type Pump, operating hours 5000 -- predict remaining useful life

Model Accuracy Testing

Validate RPT-1's prediction quality on your specific dataset. The system masks known values, runs predictions, and reports accuracy metrics so you can assess reliability before acting on results.

Example queries:

Test the model accuracy for predicting delivery_time

How accurate is the model at predicting churn?

Run an accuracy benchmark on the revenue column

Data Questions

Ask questions about your existing data without running predictions. Get aggregations, distributions, comparisons, and summaries in natural language.

Example queries:

What's the average revenue by region?

Summarize total quantity across all suppliers

Which product category has the highest churn rate?

Follow-Up Conversations

Continue the conversation with context-aware follow-ups. Reference previous results, refine scenarios, or drill deeper without repeating context.

Example queries:

Do the same but only for the EMEA region

What about Paris instead?

Why did that prediction change?

Quick Start

1. **Upload your CSV** -- drag and drop or browse for your data file
2. **Open Chat-RPT** -- click the chat icon in the bottom-right corner
3. **Ask a question** -- type any question about your data in natural language
4. **Follow up** -- refine your analysis with follow-up questions or click suggested actions

Chat API Reference

POST /api/chat

Send a natural language message about your data and receive an AI-powered response with predictions, analysis, or recommendations.

Authentication

Uses the same Bearer token authentication as `/api/predict` . Include your token in the `Authorization` header.

Request Body

```
{
  "messages": [
    {
      "role": "user",
      "content": "What if I change the supplier to Acme Corp, predict th
    }
  ],
  "tableContext": {
    "columns": ["OrderID", "Supplier", "Destination", "Quantity", "Deliv
    "rowCount": 150,
    "sampleRows": [
      {"OrderID": "ORD-001", "Supplier": "GlobalTech", "Destination": "T
      {"OrderID": "ORD-002", "Supplier": "Acme Corp", "Destination": "Be
    ],
    "columnStats": [
      {"column": "Supplier", "type": "categorical", "uniqueCount": 5},
      {"column": "Quantity", "type": "numeric", "min": 10, "max": 500, "
      {"column": "DeliveryCost", "type": "numeric", "min": 500, "max": 8
    ]
  }
}
```

Request Parameters:

Parameter	Type	Required	Description
`messages`	Array	Yes	Conversation messages. Each has `role` ("user" or "assistant") and `content` (string).
`tableContext`	Object	No	Metadata about the active dataset. Required for prediction-related queries.
`conversationContext`	Object	No	Context from the previous turn (for follow-up queries). Includes `lastIntent` and `lastQueryParams`.

Table Context Fields:

Field	Type	Description
`columns`	string[]	Array of column names from the dataset
`rowCount`	number	Total number of rows in the dataset
`sampleRows`	object[]	First few rows of data for context (up to 5)
`columnStats`	object[]	Statistical summaries per column (type, min/max/avg or uniqueCount)

Response

```
{
  "content": "I'll simulate changing the supplier to Acme Corp and predi
  "timestamp": "2026-04-29T10:30:45.123Z",
  "intent": "WHAT_IF",
  "suggestions": [
    {"label": "Try a different supplier", "query": "What if I change the
    {"label": "Check accuracy", "query": "Test the model accuracy for De
    {"label": "Reduce quantity", "query": "What if I also reduce quantit
  ]
}
```

Response Fields:

Field	Type	Always Present	Description
`content`	string	Yes	Natural language response explaining results
`timestamp`	string	Yes	ISO 8601 timestamp of the response
`intent`	string	No	Classified intent type (see table below)
`suggestions`	array	No	Contextual follow-up suggestions with `label` and `query`

Intent Types

The API classifies each message into one of these intent types:

Intent	Description	Example Query
`WHAT_IF`	Simulate data changes and predict outcomes	"What if I change supplier to X?"
`ADD_ROW`	Add a new record and predict a target value	"New order: quantity 500, predict cost"
`ACCURACY`	Test model prediction accuracy on a column	"Test accuracy for revenue"
`DATA_QUESTION`	Ask about existing data (aggregation, filtering)	"What's the average cost by region?"
`FOLLOW_UP`	Reference or refine a previous query/result	"Do the same for EMEA only"
`UNSUPPORTED`	Request outside the system's capabilities	"Submit feedback using the tile at the top"

Example: Data Question

Request:

```
{
  "messages": [
    {"role": "user", "content": "Summarize total quantity across all sup
  ],
  "tableContext": {
```

```
    "columns": ["OrderID", "Supplier", "Quantity", "DeliveryCost"],
    "rowCount": 150,
    "sampleRows": [
      {"OrderID": "ORD-001", "Supplier": "GlobalTech", "Quantity": 200,
      {"OrderID": "ORD-002", "Supplier": "Acme Corp", "Quantity": 350, "
    ],
    "columnStats": [
      {"column": "Supplier", "type": "categorical", "uniqueCount": 5},
      {"column": "Quantity", "type": "numeric", "min": 10, "max": 500, "
    ]
  }
}
```

Response:

```
{
  "content": "The total quantity across all suppliers is 27,750 units. B
  "timestamp": "2026-04-29T10:32:12.456Z",
  "intent": "DATA_QUESTION",
  "suggestions": [
    {"label": "Compare costs", "query": "What's the average delivery cos
    {"label": "Find outliers", "query": "Which orders have unusually hig
  ]
}
```

Example: Follow-Up with Context

To send a follow-up that references a previous interaction, include the conversation history and context:

```
{
  "messages": [
    {"role": "user", "content": "What if I change supplier to Acme Corp,
    {"role": "assistant", "content": "Switching to Acme Corp reduces del
    {"role": "user", "content": "Do the same but only for Berlin orders"
  ],
  "tableContext": { "..." : "..." },
  "conversationContext": {
    "lastIntent": "WHAT_IF",
    "lastQueryParams": {
```

```
      "updates": {"Supplier": "Acme Corp"},
      "predictColumn": "DeliveryCost",
      "simulationMode": true
    }
  }
}
```

Rate Limits

Chat requests have separate rate limits from the predict API:

- **Chat**: 25 requests per minute per user
- **Predict**: 30 requests per hour per user

If rate-limited, you'll receive a `429` response with a `Retry-After` header.

Error Codes

Same error codes as the predict API (400, 401, 429, 500, 503). See **API Reference** for details.

Legal

Legal Documents

Access our legal documents and policies:

- Privacy Policy: Learn how we collect, use, and protect your personal information and data.
- Terms of Service: Review our terms and conditions for using the SAP-RPT Model platform.
- Cookies Policy: Understand how we use cookies and similar technologies on our platform.

Your API Token

Use this token to authenticate API requests

Your API tokens allow you to interact with SAP-RPT Model programmatically. Manage tokens in Settings.

```
No active tokens — create one in Settings
```

Security Note: Treat this token like a password. Never commit it to version control or share it in public forums.

Using Your Token

Include your API token in the Authorization header of your requests:

```
curl -X POST https://rpt.cloud.sap/api/predict \
  -H "Authorization: Bearer No active tokens — create one in Settings" \
  -H "Content-Type: application/json" \
  -d '{"data": [...]}'
```